23 It would take 120 minutes to fill a swimming pool using water from 5 taps.

(a) How many minutes will it take to fill the pool if only 3 of the taps are used?

Taps × minutes stays constant: 5 × 120 = 600.

1 mark
Use inverse proportion: 120 × 5 ÷ 3

For 3 taps: 600 ÷ 3 = 200 minutes.

1 mark
Correct time

200 minutes

(2)

(b) State one assumption you made in working out your answer to part (a).

All taps supply water at the same constant rate.

1 mark
State the rate assumption

(1)

(Total for Question 23 is 3 marks)